

1) $A = \begin{bmatrix} -4 & 0 & 0 \\ 3 & 2 & 0 \\ -2 & 0 & 5 \end{bmatrix}$ matrisin özdeğerleri bul (1)

$$|A - \lambda I| = 0 \quad \begin{vmatrix} -4-\lambda & 0 & 0 \\ 3 & 2-\lambda & 0 \\ -2 & 0 & 5-\lambda \end{vmatrix} = -4-\lambda \begin{vmatrix} 2-\lambda & 0 \\ 0 & 5-\lambda \end{vmatrix}$$

$$\lambda_1 = -4 \quad \lambda_2 = 2 \quad \lambda_3 = 5 //$$

2) $A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 \end{bmatrix}$ matrisin özdeğerleri bul.

$\lambda_1 = 1 \quad \lambda_2 = -4 \quad \lambda_3 = 5 \quad \lambda_4 = 2 \quad \lambda_5 = -1$ Köşer matris

3) $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ matrisin özdeğerleri bul $\lambda = 5$ dir.

A^2 matrisin buna eşit özdeğerleri bul.

$$A^2 = \begin{bmatrix} 7 & 6 & 6 \\ 12 & 13 & 12 \\ 6 & 6 & 7 \end{bmatrix} \quad \lambda = 5^2 = 25 //$$

$$Ax = \lambda x \quad A \cdot Ax = \lambda \underbrace{Ax}_{\lambda x} \quad \checkmark$$

$$\boxed{A^2 x = \lambda^2 x}$$

4) $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ -1 & -2 & 2 \end{bmatrix}$ matrisin karakteristik denklemleri.

5)

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0.$$

$$A^3 - 6A^2 + 11A - 6I = 0.$$

$$! A^3 = A^2 \cdot A = \begin{bmatrix} 2 & -12 & 13 \\ 6 & 20 & -13 \\ -13 & -26 & 14 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & -2 & 4 \\ 1 & 6 & -4 \\ -4 & -8 & 5 \end{bmatrix}$$

Not $B = Q^{-1} A Q$ olarak seçildi ters alınırsa bir Q matrisi varsa A ve B benzerdir.
 $|A| = |B|$ dir.
 $\text{tr} A = \text{tr} B$

12. soru.

25) A kare matrisin köşegenleştirilen Q ve k pozitif bir tam sayı olsun

$$Q^{-1} A Q = B \quad A^k = Q B^k Q^{-1}$$

$$k=2 \quad B^2 = (Q^{-1} A Q)^2 = Q^{-1} A Q Q^{-1} A Q = Q^{-1} A^2 Q$$

$$A^2 = Q B^2 Q^{-1}$$

$$\boxed{A^k = Q B^k Q^{-1}}$$

5)

$$A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

$$A^7 = ?$$

$$Q = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 0 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$Q^{-1} = \frac{1}{6} \begin{bmatrix} -3 & 0 & 3 \\ 1 & 2 & -2 \\ 2 & -2 & -1 \end{bmatrix}$$

$$B = Q^{-1} A Q = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$B^7 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 128 \end{bmatrix}$$

$$A^7 = -\frac{1}{6} \begin{bmatrix} -3 & 0 & 3 \\ 1 & 2 & -2 \\ 2 & -2 & -2 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 128 \end{bmatrix}$$

A)

$$\begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}$$

symmetrische matrix, hermitisch.

Eigenwerte bestimmen

$$|A - \lambda I| = (\lambda - 3)^2 \cdot \lambda = 0 \quad \lambda_{1,2} = 3 \quad \lambda_3 = 0$$

$$\lambda = 3 \rightarrow \begin{pmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} x_2 = t \\ x_3 = s \\ x_1 = -s - t \end{array}$$

$$X = \begin{pmatrix} -s-t \\ t \\ s \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} s + \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} t \rightarrow \left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} \right\}$$

$$\lambda = 0 \rightarrow \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} x_3 = t \\ x_2 = t \\ x_1 = t \end{array} \quad X = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\left\{ \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} \quad \text{Spaltenvektoren bilden orthonormale Basis}$$

$$v_1 = x_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$v_2 = x_2 - \frac{\langle x_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/2 \\ -1/2 \\ 1 \end{pmatrix}$$

$$v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - 0 - 0 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} -1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & 2/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}$$

$$S = \begin{pmatrix} -1 & -1/2 & 1 \\ 1 & -1/2 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

$$Q^T A Q = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} //$$

ODEV

$$\begin{pmatrix} 1 & -4 & 2 \\ -4 & 1 & -2 \\ 2 & -2 & -2 \end{pmatrix}$$

symmetrische Matrix

Eigenwerte

$$Q = \begin{pmatrix} 1/\sqrt{2} & -\sqrt{2}/6 \\ 1/\sqrt{2} & \sqrt{2}/6 \\ 0 & 2/\sqrt{3} \end{pmatrix}$$

$$Q^T A Q$$

$$= \begin{pmatrix} -3 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{pmatrix} //$$

ÖDEÜ (17)

$$A = \begin{bmatrix} 5 & 0 \\ 2 & -3 \end{bmatrix}, \quad \begin{bmatrix} -2 & 1 \\ 5 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & -4 \\ 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} -1 & 3 \\ 5 & 1 \end{bmatrix}$$

$\lambda_1 = 5, \lambda_2 = -3$ $\lambda_1 = -3, \lambda_2 = 3$ $\lambda_1 = 1, \lambda_2 = 2$ $\lambda_1 = -4, \lambda_2 = 4$
 özdeğer ve özvektörler bulma
 $\downarrow$ $\downarrow$ $\downarrow$ $\downarrow$
 $\begin{bmatrix} 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ 1 \end{bmatrix}$ $\begin{bmatrix} -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3/5 \\ 1 \end{bmatrix}$

b) $A_{2 \times 2}$ matrisin karakteristik denklemini (YAPMA)
 $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$ old. gösterin.

$$\lambda^2 - (a+d)\lambda + ad - bc = 0.$$

(6)

ÖDEÜ

$$A_{3 \times 3} = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$\lambda^3 - \text{tr}(A)\lambda^2 + \left(\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix} + \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \right) \cdot \lambda - |A| = 0$$

old. gösterin.



↓
 Anxn matrisin karakteristik denklemini
 Sabit terim $(-1)^n \det A$ olduğuna göre - (3)

a) $P(\lambda) = \lambda^3 + 7\lambda^2 + 4\lambda + 15$ old. göre $\det(A) = ?$

$n=3$ sbt term = 15

$(-1)^3 |A| = 15 \quad |A| = -15$

b) $P(\lambda) = \lambda^2 - 5\lambda + 9$

$(-1)^2 |A| = 9$

$|A| = 9$

c) $P(\lambda) = \lambda^5 - 2\lambda^4 + \lambda^3 + 7\lambda^2 - \lambda - 17$

$n=5$

$-17 = (-1)^5 \cdot |A| \quad |A| = \underline{\underline{17}}$

ÖDEV (21)

$A = \begin{bmatrix} -1 & 1 & -1 \\ -2 & 2 & -1 \\ -2 & 1 & 0 \end{bmatrix}$

matrisin özdeğer ve özvektörlerini bulun -

$|A - \lambda I| = 0 \Rightarrow (\lambda - 1)^2 (\lambda + 1) = 0 \quad \lambda = 1 \quad \lambda = -1$

$\lambda = 1 \quad \begin{bmatrix} 2 & -1 & 1 \\ 2 & -1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow X = \begin{bmatrix} \frac{1}{2}t - \frac{1}{2}s \\ t \\ s \end{bmatrix}$

$\rightarrow \left\{ \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{pmatrix} \right\}$



$$r = -1 \rightarrow \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(4)

ÖÜÜ

(8) $A = \begin{bmatrix} 1 & m \\ 1 & 1 \end{bmatrix}$ Matrizen besitzen gleiche reelle
Eigenwerte oder von m abhängig?

$$\lambda^2 - \text{tr}(A) \cdot \lambda + \det(A) = 0 \quad \lambda^2 - 2\lambda + (1-m) = 0$$

$$\Delta = 4 - 4(1-m) > 0.$$

$m > 0$. d.h.

ÖÜÜ

(16) $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ -3 & 5 & 0 \end{bmatrix}$ Matrix Eigenwertbestimmung.

Charakteristisches Polynom: $(\lambda - 1)(\lambda - 2)^2 = 0$

$$\begin{pmatrix} -1 \\ 1 \\ -8 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Hauptachsentransformation

gibt es.

(9)

7) $A_{2 \times 2}$ Matrizen Eigenwerte $\lambda_1 = 2$ und $\lambda_2 = 5$

$$|A| = ?$$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$10 = (-1)^2 \cdot |A|$$

$|A| = 10$ //

B.) $\textcircled{15}$ $A = \begin{bmatrix} 4 & 0 & -2 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

Matrix ist regulär.

$$|A - \lambda I| = 0 \quad (\lambda - 3)(\lambda - 1)(\lambda - 2) = 0$$

$$\lambda_1 = 3 \rightarrow \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad \lambda_2 = 1 \rightarrow \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3 = 2 \rightarrow \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

(Sinhelk dpol)

$$Q = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad Q^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 2 \end{bmatrix} \quad Q^{-1}AQ = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

~~Özet~~ (Göz) $\textcircled{22}$

$$A = \begin{bmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{bmatrix} \text{ regulär}$$

$$\begin{aligned} \lambda_1 &= 9 \rightarrow \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \\ \lambda_2 &= 3 \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ \lambda_3 &= -3 \rightarrow \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \end{aligned}$$

Sinhelk

Simultane Diagonalisierung
orthogonal Q

$$\begin{bmatrix} -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$

$$Q^{-1}AQ = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$\begin{bmatrix} -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{bmatrix} \begin{bmatrix} -2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix} =$$

$$\begin{bmatrix} -18/\sqrt{6} & 3/\sqrt{3} & 0 \\ 9/\sqrt{6} & 3/\sqrt{3} & 3/\sqrt{2} \\ 9/\sqrt{6} & 3/\sqrt{3} & -3/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

(5)

$A = \begin{bmatrix} 1 & 8 \\ 4 & 5 \end{bmatrix}$ matrisi veribyer A^n matrisini bulur.

(26)

$$Q \rightarrow \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \quad Q^{-1} = -\frac{1}{3} \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$$

$$D = Q^{-1} A Q = \begin{bmatrix} -3 & 0 \\ 0 & 9 \end{bmatrix}$$

$$A^n = Q D^n Q^{-1} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} (-3)^n & 0 \\ 0 & 9^n \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & -2 \end{bmatrix}$$

$$A^n = -\frac{1}{3} \begin{bmatrix} -2(-3)^n - 9^n & 2(-3)^n - 2(9)^n \\ (-3)^n - 9^n & -(-3)^n - 2(9^n) \end{bmatrix} //$$

(27)

$A = \begin{bmatrix} 1 & 8 \\ 4 & 5 \end{bmatrix}$ matrisi veribyer Cayley-Hamilton yordamıyla A^4 ve A^{-2} bulur.

$$|A - \lambda I| = 0 \quad \lambda^2 - 6\lambda - 27 = 0$$

$$A^2 - 6A - 27I = 0$$

$$A^2 = 6A - 27I$$

$$A^3 = 6A^2 - 27A$$

$$= 6(6A - 27I) - 27A$$

$$= 36A - 162I - 27A$$

$$= 9A - 162I$$

$$A^4 = 9A^2 - 162A$$

$$= 9(6A - 27I) - 162A$$

$$= 54A - 243I - 162A$$

$$A^4 = 6A \cdot A^2 - 27 \cdot A^2$$

$$= 6A(6A - 27I) - 27(6A - 27I)$$

$$= 36A^2 + 324A + 729I$$

$$= 36(6A - 27I) + 324A + 729I$$

$$= 540A - 1701I$$

$$A^4 = \begin{bmatrix} 224 & 1 & 4320 \\ 2160 & 4401 \end{bmatrix}$$

$$A^{-1} \rightarrow$$

$$A^2 - 6A - 27I = 0$$

$$A - 6I - 27A^{-1} = 0$$

$$A^{-1} = \frac{A - 6I}{27} = \frac{1}{27} \begin{bmatrix} -5 & 8 \\ 4 & -1 \end{bmatrix}$$

↓ 3DEU ver.

ÖDEV (12) Cayley Hamilton yordamı

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ -1 & -2 & 2 \end{bmatrix} \text{ matrisin inverso bul.}$$

$$\text{Solukn. } \frac{1}{6} \begin{bmatrix} 2 & -2 & -2 \\ 1 & 5 & 2 \\ 2 & 4 & 4 \end{bmatrix}$$

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